

Zero-Post-Processing Micro-Transfer-Printed Heterogeneous Si/TFLN Mach-Zehnder Modulators with >67-GHz Bandwidth

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Abstract We micro-transfer-print fully pre-fabricated thin-film lithium niobate modulators, incorporating RF CPW electrodes and SiO₂ cladding, onto a silicon platform using a Mo sacrificial layer. Eliminating post-printing fabrication, the 3-mm-long device achieves a 3-dB bandwidth exceeding 67 GHz, enabling scalable integration. ©2026 The Author(s)

Introduction

Generative AI advancements are driving data center interconnect capacities toward 3.2 Tb/s. While silicon photonics (SiPh) provides a mature integration platform, conventional Si modulators face intrinsic bandwidth-efficiency trade-offs. Thin-film lithium niobate (TFLN) overcomes this bottleneck, offering ultra-fast, energy-efficient electro-optic (EO) modulation [1].

To heterogeneously integrate TFLN onto SiPh, micro-transfer printing (μ -TP) provides a highly scalable solution [2, 3]. Unlike conventional die-to-wafer bonding [4], which consumes a large chip footprint and restricts the co-integration of multiple disparate materials, μ -TP enables the high-density, flexible assembly of diverse photonic components. Furthermore, it avoids CMOS lithium contamination by allowing integration after back-end-of-line (BEOL) processes. Recently, we demonstrated the μ -TP of a 1.1-cm-long TFLN Mach-Zehnder modulator (MZM) with a half-wave voltage of 2.7 V [5], and heterogeneous Si/TFLN MZM with <0.1-dB transition loss and >67-GHz EO bandwidth via optimized interlayer couplers [6]. However, existing μ -TP methods still require costly post-

printing fabrication on the target chip, such as SiO₂ over-cladding and RF electrode formation.

In this paper, we demonstrate the μ -TP integration of a fully pre-fabricated TFLN coupon onto a SiPh platform and zero-post-processing heterogeneous Si/TFLN MZM. In conventional TFLN μ -TP, the SiO₂ buried oxide (BOX) layer of a commercial LNOI substrate serves as the sacrificial layer [5,6]. However, for device coupons incorporating pre-fabricated electrodes, removing the BOX layer to release the TFLN typically causes damage to the SiO₂ buffer layer between the electrodes and the LN waveguides, making the standard approach unsuitable. To address this challenge, we proposed using a molybdenum (Mo) sacrificial layer instead of conventional SiO₂. Unlike conventional bonding via CMP-polished SiO₂, our approach employs a locally formed deeply etched window (DW) to directly bond the coupon onto the exposed Si layer. By realizing a zero-post-processing heterogeneous modulator, this approach ensures strict CMOS back-end-of-line (BEOL) compatibility. This localized DW integration enables the dense co-integration of multiple disparate materials on a single platform, while realizing high-speed optical modulation.

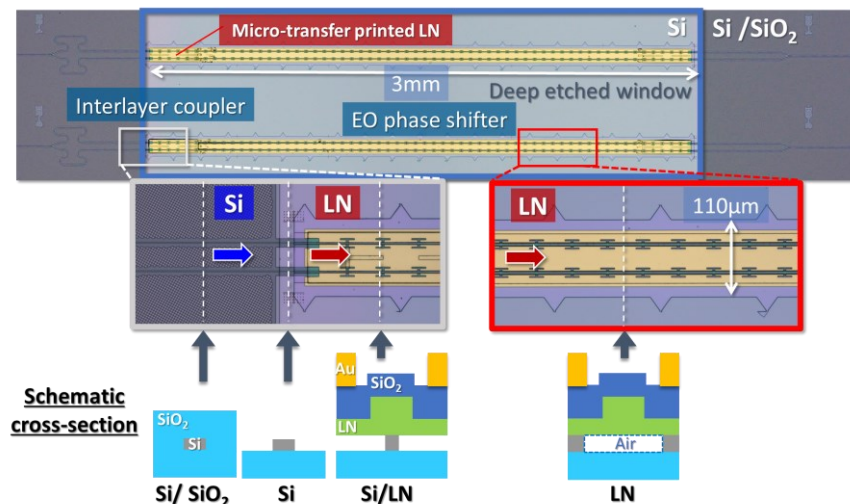


Fig. 1: Microscopic image of heterogeneous Si/LN hybrid MZM and schematic cross-sectional views of each device section.

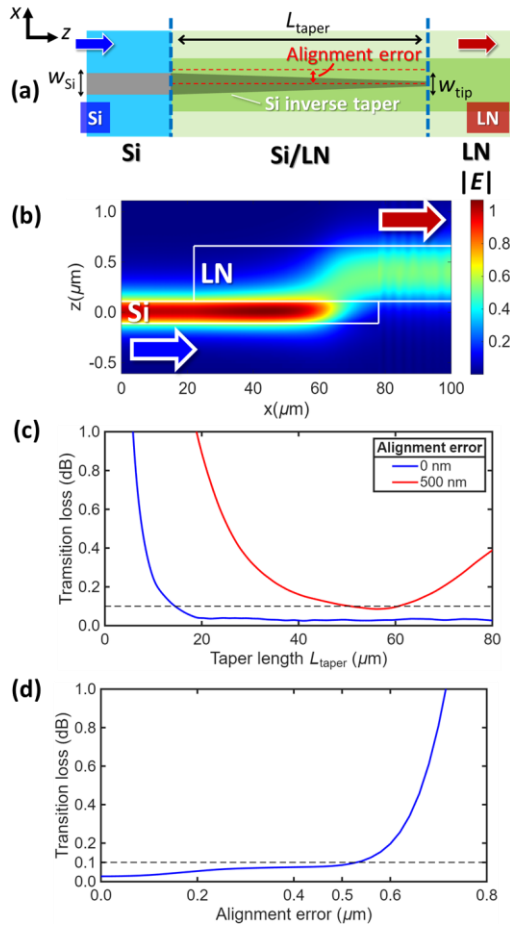


Fig. 2: (a) Schematic diagrams of top view of the inter-layer coupler. (b) Color map of normalized electric field distribution of light propagating from Si waveguide to LN waveguide. Simulated transition loss of inter-layer coupler as a function of (c) taper length and (d) alignment errors.

Device Design and Fabrication

Figure 1 shows a microscopic image of the fabricated heterogeneous Si/LN MZM, consisting of a Si MMI-based splitter/combiner and a TFLN coupon incorporating two parallel etched LN waveguides and GSG coplanar waveguide (CPW) electrodes. Input light is equally split by a Si MMI and completely transferred from the Si waveguide to the LN waveguide via inverse-taper interlayer couplers (Fig. 2(a)). The light in the LN waveguides is modulated in a push-pull configuration, coupled back to the Si waveguides through the interlayer couplers, and recombined by an output Si MMI.

An interlayer coupler with high alignment tolerance and low loss was designed by optimizing the tip width w_{tip} and taper length L_{taper} of the Si inverse taper. Figure 2(c) shows the calculated loss as a function of taper length for a Si inverse taper with a tip width of 200 nm, under alignment errors of 0 nm and 500 nm (alignment error is defined as the positional deviation of the waveguide

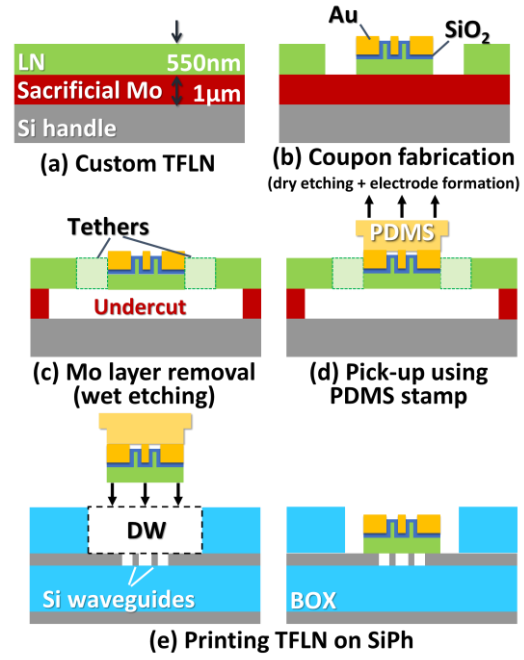


Fig. 3: Schematic cross-sectional illustrations of the fabrication process

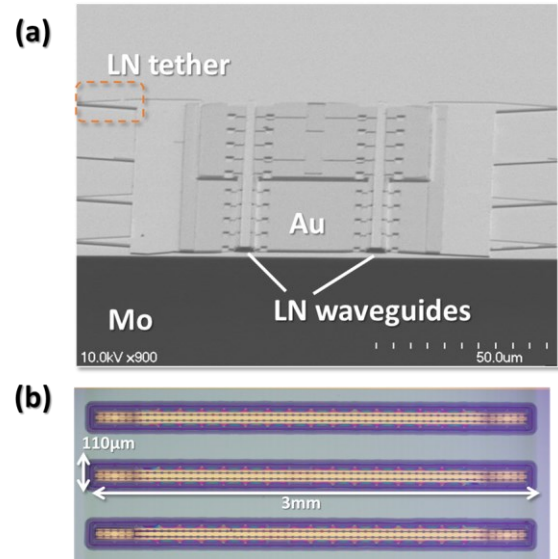


Fig. 4: (a) SEM image of a fully pre-fabricated TFLN coupon. (b) suspended TFLN coupons after removal of Mo sacrificial layer

centers). Without alignment error, the loss converges beyond 20 μm , whereas with a 500-nm alignment error, it is minimized at a taper length between 50 and 60 μm . Therefore, a taper length of 56 μm was chosen. Interlayer transition through a SiO₂ layer requires a taper length of approximately 300 μm [6], whereas direct transfer from Si to LN without an intervening SiO₂ layer enables a substantial reduction to less than one-fifth of this length. Figure 2(d) shows the transition loss as a function of alignment error for a taper length of 50 μm . For a 3σ alignment error within 500 nm in our bonding system, the

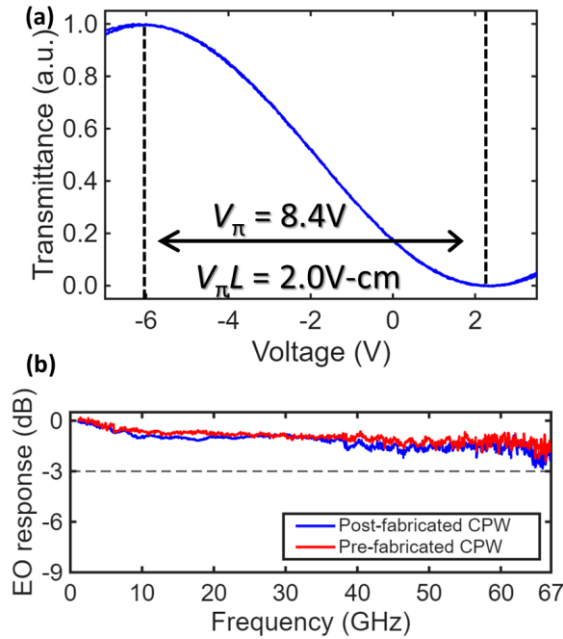


Fig. 5: (a) Normalized transmittance as a function of the applied voltage and (b) EO response of heterogenous Si/TFLN

transition loss per Si–LN waveguide interface remains below 0.1 dB.

The fabrication process is as follows. First, a custom-made 4-inch wafer with an embedded Mo sacrificial layer and a 550-nm-thick TFLN layer was fabricated on a Si substrate (Fig. 3(a)). Then, fully pre-fabricated LN device coupons were formed by dry etching the LN layer, followed by SiO_2 buffer deposition and electrode formation (Fig. 3(b)). An SEM image of a fabricated coupon is shown in Fig. 4(a). Subsequently, the Mo sacrificial layer was selectively removed using an acidic etchant solution, resulting in a suspended structure in which the coupon is supported by triangular monolithic LN tether structures. Microscopic images of several suspended coupons are shown in Fig. 4(b). Next, using μ -TP, the suspended LN coupon was picked up with a PDMS elastomer stamp (Fig. 3(d)) and printed under applied pressure onto a deeply etched window (DW), where the Si waveguide is exposed (Fig. 3(e)). This bonding is based on hydrophilic interactions and provides strong adhesion; no additional adhesive, such as BCB, is required.

In heterogeneous integration, bonding via CMP-polished SiO_2 is commonly employed to ensure planarity in the integration region. However, this approach results in a uniformly thinned overcladding layer across the wafer, which can increase coupling loss during edge coupling. In addition, it is difficult to perform CMP after integrating metals such as heaters on top of the waveguide. In contrast, the DW approach enables fabrication after forming electrical interconnects while maintaining sufficient overcladding

thickness at the chip facet. Furthermore, unlike previously reported TFLN μ -TP integration methods, our approach eliminates the need for post-processing steps, such as waveguide etching or electrical interconnect formation after the μ -TP process, thereby enabling full compatibility with CMOS BEOL integration.

Characterization and Discussion

To measure the half-wave voltage V_{π} of the heterogeneous MZM, a triangular voltage was applied in a push–pull configuration through the CPW at a repetition rate of 100 kHz, and the optical output was detected using a photodetector and monitored on a real-time oscilloscope. The transmittance as a function of the applied voltage is shown in Fig. 5(a), from which V_{π} was estimated to be 8.4 V. For the 2.4-mm-long phase shifters in the push–pull configuration, the half-wave voltage–length product $V_{\pi}L$ was 2.0 V·cm, which is in good agreement with the simulated value.

Subsequently, the high-speed EO response of the heterogeneous MZM was characterized. The optical output from the MZM was detected by a high-speed PD and subsequently analyzed using a vector network analyzer. The red curve in Fig. 5(b) shows the measured EO bandwidth of the MZM, indicating a 3-dB EO bandwidth of over 67 GHz. Figure 5(b) also shows the EO characteristics of a device in which the CPW was formed after the TFLN transfer [6], for comparison. It can be seen that there is no significant difference in the EO response compared to the device with pre-fabricated CPWs.

Conclusions

In summary, the micro-transfer printing of fully pre-fabricated TFLN modulators utilizing a molybdenum sacrificial layer marks a critical advancement in heterogeneous silicon photonics. By locally bonding the device coupons into deeply etched window, this approach eliminates complex post-printing fabrication and ensures strict CMOS BEOL compatibility. More importantly, this localized DW integration highlights the unique advantage of μ -TP: enabling the dense co-integration of multiple disparate materials on a single platform. Operating with a bandwidth exceeding 67 GHz, this zero-post-processing strategy maintains excellent high-speed performance, paving the way for next-generation multi-material 3.2 Tb/s optical interconnects.

Acknowledgements

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